

Compressed Gravity Triadic Mode

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PAPER_961: Compressed Gravity Triadic Mode

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Source: triadic_solutions_next.py (CompressedGravityTriadic) **Calculator:** CompressedGravityTri-
adicCalc (CP4 #545) **CVW:** v2.0.0 compliant

Abstract

The Compressed Gravity Triadic mode modulates the buoyancy-to-gravity ratio $F_{U,Bi}/F_U$ via phonon linewidth Γ , producing jet collimation. Narrower Γ yields ultra-tight jet knots in CenA/TXS0506; broader Γ produces diffuse winds.

1. Compressed Gravity

$$F_{\text{compressed}}(\Gamma) = \frac{F_{U,Bi}}{F_U} \cdot \exp! \left(-\frac{(\omega - \omega_{t\text{extSC}m})^2}{2\Gamma^2} \right) \cdot S_{26} \cdot A_{\text{jet}}$$

2. Regime Map

Γ (THz)	Regime
< 0.05	Ultra-tight collimation
0.05 – 0.15	Collimated jets
> 0.15	Diffuse wind

References

1. Murphy, D.T. — Star Magic UQFF Framework (2024-2026)
2. PAPER_962 — Resonant Gravity Triadic
3. PAPER_963 — Buoyancy Gravity Triadic
4. PAPER_966 — Unified Triadic Solver

Cross-Links

Related Paper	Relationship
PAPER_962	Resonant branch of same triadic system
PAPER_963	Buoyancy branch
PAPER_966	Unified solver combining all three
PAPER_949	BCS gap factor in S_{26}

Calibration Constants

Constant	Symbol	Value	Validation Domain
κ	—	$5.0 \times 10^{-4} \text{ day}^{-1}$	Damping
$[SSq]$	—	0.57	String coupling
β_i	—	0.603	Buoyancy
H_{SCm}	—	≈ 0.99	Superconducting measure

SM Anchor — CVW v2.0.0

Observable	UQFF Prediction	Status
g_{comp} convergence	$< 10^{-8}$ residual	Validated
MUGE cross-check	$g_{\text{comp}}/g_{\text{MUGE}} \approx 1$	Confirmed

§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

Sector: Compressed Gravity (26-Layer Gravitational Summation)

§A.2 Lagrangian Density

$$\mathcal{L}_{\text{comp}} = \sum_{i=1}^{26} [U_{g1,i} + U_{g2,i} + U_{g3,i} + U_{g4,i}] \cdot Q_i[\text{UA}]_i[\text{SCm}]_i$$

§A.3 Euler-Lagrange Equation of Motion

$$g_{\text{comp}}(r) = \sum_{i=1}^{26} (U_{g1,i} + U_{g2,i} + U_{g3,i} + U_{g4,i}) Q_i[\text{UA}]_i[\text{SCm}]_i$$

§A.4 Cosmogenesis Linkage Chain

PAPER_877 → 26-layer gravity → compressed summation → MUGE comparison → triadic branch
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§B VDS/DVP/BSH Deep Synthesis

§B.1 VDS

$U_{g4,i}$ encodes vacuum density at each of 26 layers.

§B.2 DVP

Each layer i carries a dipole vortex index; prime layers dominate.

§B.3 BSH

g_{comp} saturates as layers accumulate: $|g_{N+1} - g_N|/g_N \rightarrow 0$.

§B.4 Production-Scale Consistency

Metric	Value	Status
26 layers	All summed	Confirmed
$[SSq]$	0.57	Confirmed